Sticks

In a random experiment, we break a one-foot stick “at a random place.” We model this experiment using a random variable X representing the distance (in feet) from the left end of the stick where we break it. We take X to have the uniform distribution on [0, 1]. The density function f for this distribution is defined by $f(x) = 0$, if $x < 0$ or $x > 1$ and $f(x) = 1$ if $0 \le x \le 1$. The point where we break the stick is equally likely to be any place between 0 and 1 feet from the left end.

a. Consider the event “the smaller piece is less than 0.25 feet long.” What values of X correspond to this event? To answer, draw a picture of the sick, and shade the places where you could break it so that the smaller piece is less than 0.25 feet long.

b. What is the probability that the smaller piece is less than 0.25 feet long? To answer this question, calculate what fraction of the stick you shaded.

c. For a number x between 0 and 1, what is the probability $F(x)$ that the smaller piece of the stick will be less than x feet long? Sketch a picture again to work this out.

d. The function F that you calculated is the distribution function for the random variable Y representing the length of the shorter piece of the stick. To find the density function, calculate F'. Do you recognize the distribution of Y?